

Joy Sarker

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RESEARCH INTERESTS

AI coding agents: why they sometimes hallucinate, give wrong information or go around in circles without making progress, and how to catch these mistakes.

EDUCATION

State University of Bangladesh (SUB)

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

2018 – 2022

- CGPA: **3.96 / 4.00**; first in class (highest CGPA in the graduating class)
- Dean's Award (Silver Medal), 6th Convocation, 2022; Merit Scholarship in every semester
- Relevant coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Algorithm Analysis, Theory of Computation, Discrete Mathematics, Software Engineering

PROFESSIONAL EXPERIENCE

Therap (BD) Ltd.

Dhaka, Bangladesh

Software Engineer II (Associate SE → SE → SE II)

June 2022 – Present

Bangladesh affiliate of Therap Services (US), which makes software for human services providers.

- Develop features with Spring MVC, JSP and JavaScript from business requirement documents.
- Since June 2025, write JUnit unit tests and Playwright functional tests for each feature.
- Built a summary report with a documented API, an organizational chart, and a review feature with comments.
- Fix security issues and write a test that confirms each fix.
- Rewrote slow Hibernate (HQL) queries to use composite indexes and added missing indexes on large tables.
- Moved critical business logic from the client to the server to improve security and data integrity.
- Implemented optimistic locking on Hibernate entities to prevent conflicting concurrent edits.
- Review code for my own team and for other teams in the company.
- Mentored 5+ junior developers in Spring and JPA, and guided three newly hired engineers through their training.

Stack: Java, Spring MVC, Hibernate, Oracle SQL, JSP, JavaScript, XSLT, JUnit, Playwright, Gradle

ACADEMIC PROJECTS

Algorithm & Data Structure Visualization Engine [\[code\]](#) [\[demo\]](#)

2022

B.Sc. final-year project, with M. Huq and K. Somapriya

Web app that animates sorting algorithms (bubble, merge and quick sort) and graph algorithms (Dijkstra, BFS, DFS) step by step. Users can enter custom inputs to compare best-case and worst-case behavior.

Decision Tree and k-NN Classification (Machine Learning course project) [\[code\]](#) [\[report\]](#)

2021

Trained a decision tree and a k-nearest neighbors classifier with scikit-learn on a small categorical dataset. Encoded the features, plotted the tree and compared k = 1 to 5.

HONORS & AWARDS

Dean's Award (Silver Medal), State University of Bangladesh

2022

Honorable Mention, ICPC Asia Dhaka Regional ([ICPC record](#))

2019, 2020, 2021

Reached Round 2, Facebook Hacker Cup (top 8% of 35,000+ participants)

2021

Qualified for Round 1, Google Code Jam

2022

Codeforces Specialist ([PrimeX](#)); solved 2,000+ problems on online judges

VOLUNTEER TEACHING & SERVICE

Competitive Programming Trainer, State University of Bangladesh

Mar 2020 – Aug 2022

Trained younger students for programming contests.

Problem Setter and Judge, ICPC-style contests on Toph, Dept. of CSE, SUB

2021 – 2024

Wrote more than 10 problems, each with its statement, reference solution and test data, for four contests, including [SUB CSE Tech Carnival 2024](#) and [SUB CSE FEST IDPC 2023](#).

Invited Speaker, Code Quest Workshop on Competitive Programming, Dept. of CSE, SUB

Aug 2023

President, SUB Computing Club

Sep 2021 – Jun 2022

Organized programming contests, seminars and workshops for the CSE department.

SKILLS

Languages	Python, Java, C/C++, JavaScript
Frameworks	Spring MVC, Spring Boot, JPA/Hibernate, JSP/Servlets
Testing	JUnit, Playwright
Databases	Oracle SQL, MySQL, JDBC
Tools	Git, Gradle